SG No. 214 O^8 $I4_132$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$
* Wyckoff site: `8a`, site symmetry: `.32`

Table 1: Wyckoff bond: `8a@8a`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,5,9,-14,-19,-24]`
2	$[-X,\,-X,\,X]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[2,7,12,-13,-17,-21]`
3	$[-X,\,X,\,-X]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X,\,-X,\,-X]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[4,6,11,-16,-18,-23]`

Table 2: Wyckoff bond: `24b@8a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\,X,\,Y]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,-14]`
2	$[-X,\,-X,\,Y]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[2,-13]`
3	$[-X,\,X,\,-Y]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[3,-15]`
4	$[X,\,-X,\,-Y]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[4,-16]`
5	$[Y,\,X,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[5,-24]`
6	$[Y,\,-X,\,-X]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[6,-23]`
7	$[-Y,\,-X,\,X]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[7,-21]`
8	$[-Y,\,X,\,-X]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[8,-22]`
9	$[X,\,Y,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[9,-19]`
10	$[-X,\,Y,\,-X]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[10,-20]`
11	$[X,\,-Y,\,-X]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[11,-18]`
12	$[-X,\,-Y,\,X]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[12,-17]`

Table 3: Wyckoff bond: `24c@8a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\,-X,\,0]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[1,14]`
2	$[-X,\,X,\,0]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[2,13]`
3	$[-X,\,-X,\,0]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[3,15]`
4	$[X,\,X,\,0]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[4,16]`
5	$[0,\,X,\,-X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[5,24]`
6	$[0,\,-X,\,X]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[6,23]`
7	$[0,\,-X,\,-X]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[7,21]`
8	$[0,\,X,\,X]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[8,22]`
9	$[-X,\,0,\,X]$	$\left[\frac{1}{8},\,\frac{1}{8},\,\frac{1}{8}\right]$	`[9,19]`
10	$[X,\,0,\,-X]$	$\left[\frac{7}{8},\,\frac{5}{8},\,\frac{3}{8}\right]$	`[10,20]`
11	$[-X,\,0,\,-X]$	$\left[\frac{5}{8},\,\frac{3}{8},\,\frac{7}{8}\right]$	`[11,18]`
12	$[X,\,0,\,X]$	$\left[\frac{3}{8},\,\frac{7}{8},\,\frac{5}{8}\right]$	`[12,17]`

Table 4: Wyckoff bond: `48d@8a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{5}{8}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{7}{8}, \frac{5}{8}, \frac{3}{8}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{5}{8}, \frac{3}{8}, \frac{7}{8}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[24]`

* Wyckoff site: `8b`, site symmetry: `.32`

Table 5: Wyckoff bond: `8a@8b`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1,5,9,-14,-19,-24]`
2	$[-X, -X, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2,7,12,-13,-17,-21]`
3	$[-X, X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4,6,11,-16,-18,-23]`

Table 6: Wyckoff bond: `24b@8b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1,-14]`
2	$[-X, -X, Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2,-13]`
3	$[-X, X, -Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3,-15]`
4	$[X, -X, -Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4,-16]`
5	$[Y, X, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[5,-24]`
6	$[Y, -X, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[6,-23]`
7	$[-Y, -X, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[7,-21]`

continued ...

Table 6

No.	vector	center	mapping
8	$[-Y, X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[8,-22]`
9	$[X, Y, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[9,-19]`
10	$[-X, Y, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[10,-20]`
11	$[X, -Y, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[11,-18]`
12	$[-X, -Y, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[12,-17]`

Table 7: Wyckoff bond: `24c@8b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1,14]`
2	$[-X, X, 0]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2,13]`
3	$[-X, -X, 0]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3,15]`
4	$[X, X, 0]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4,16]`
5	$[0, X, -X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[5,24]`
6	$[0, -X, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[6,23]`
7	$[0, -X, -X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[7,21]`
8	$[0, X, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[8,22]`
9	$[-X, 0, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[9,19]`
10	$[X, 0, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[10,20]`
11	$[-X, 0, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[11,18]`
12	$[X, 0, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[12,17]`

Table 8: Wyckoff bond: `48d@8b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}]$	`[19]`

continued ...

Table 8

No.	vector	center	mapping
20	$[X, -Z, Y]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[24]

* Wyckoff site: 12c, site symmetry: 2.22

Table 9: Wyckoff bond: 12a@12c, bond symmetry: 2.22

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[1,-4,18,-19]
2	$[0, -X, X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	[2,-3,-17,20]
3	$[X, 0, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	[5,-8,-14,15]
4	$[X, 0, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	[6,-7,13,-16]
5	$[X, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[9,-12,21,-24]
6	$[-X, X, 0]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[10,-11,-22,23]

Table 10: Wyckoff bond: 12b@12c, bond symmetry: 2.22

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[1,-4,-18,19]
2	$[0, -X, -X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	[2,-3,17,-20]
3	$[-X, 0, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	[5,-8,14,-15]
4	$[-X, 0, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	[6,-7,-13,16]
5	$[X, -X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[9,-12,-21,24]
6	$[-X, -X, 0]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[10,-11,22,-23]

Table 11: Wyckoff bond: 12c@12c, bond symmetry: 2.22

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[1,4,-18,-19]
2	$[-X, 0, 0]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	[2,3,-17,-20]
3	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	[5,8,-14,-15]
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	[6,7,-13,-16]
5	$[0, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	[9,12,-21,-24]
6	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	[10,11,-22,-23]

Table 12: Wyckoff bond: 24d@12c, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	[1,-19]
2	$[-Y, -X, X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	[2,-17]

continued ...

Table 12

No.	vector	center	mapping
3	$[-Y, X, -X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[12,-21]`

Table 13: Wyckoff bond: `24e@12c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[1,-18]`
2	$[-Y, -X, -X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[2,-20]`
3	$[-Y, X, X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[3,-17]`
4	$[Y, -X, X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[4,-19]`
5	$[-X, Y, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[5,-15]`
6	$[-X, -Y, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[6,-13]`
7	$[X, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[7,-16]`
8	$[X, Y, -X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[8,-14]`
9	$[X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[9,-21]`
10	$[-X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[10,-23]`
11	$[X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[11,-22]`
12	$[-X, X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[12,-24]`

Table 14: Wyckoff bond: `24f@12c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[22,-23]`

Table 15: Wyckoff bond: `48g@12c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[8]`
9	$[Y, Z, X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{8}, 0\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, \frac{3}{8}, 0\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{8}, 0, \frac{1}{4}\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[\frac{3}{8}, 0, \frac{3}{4}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[24]`

* Wyckoff site: `12d`, site symmetry: `2.22`

Table 16: Wyckoff bond: `12a@12d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[1,-4,18,-19]`
2	$[0, -X, X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[2,-3,-17,20]`
3	$[X, 0, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[5,-8,-14,15]`
4	$[X, 0, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[6,-7,13,-16]`
5	$[X, X, 0]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[10,-11,-22,23]`

Table 17: Wyckoff bond: `12b@12d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[1,-4,-18,19]`
2	$[0, -X, -X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[2,-3,17,-20]`
3	$[-X, 0, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[5,-8,14,-15]`
4	$[-X, 0, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[6,-7,-13,16]`
5	$[X, -X, 0]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[9,-12,-21,24]`

continued ...

Table 17

No.	vector	center	mapping
6	$[-X, -X, 0]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[10,-11,22,-23]`

Table 18: Wyckoff bond: `12c@12d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[1,4,-18,-19]`
2	$[-X, 0, 0]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[2,3,-17,-20]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[5,8,-14,-15]`
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[6,7,-13,-16]`
5	$[0, 0, X]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[9,12,-21,-24]`
6	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[10,11,-22,-23]`

Table 19: Wyckoff bond: `24d@12d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[1,-19]`
2	$[-Y, -X, X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[2,-17]`
3	$[-Y, X, -X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[12,-21]`

Table 20: Wyckoff bond: `24e@12d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[1,-18]`
2	$[-Y, -X, -X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[2,-20]`
3	$[-Y, X, X]$	$\left[\frac{7}{8}, 0, \frac{3}{4}\right]$	`[3,-17]`
4	$[Y, -X, X]$	$\left[\frac{5}{8}, 0, \frac{1}{4}\right]$	`[4,-19]`
5	$[-X, Y, X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[5,-15]`
6	$[-X, -Y, -X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[6,-13]`
7	$[X, -Y, X]$	$\left[\frac{3}{4}, \frac{7}{8}, 0\right]$	`[7,-16]`
8	$[X, Y, -X]$	$\left[\frac{1}{4}, \frac{5}{8}, 0\right]$	`[8,-14]`
9	$[X, -X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[9,-21]`
10	$[-X, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[10,-23]`
11	$[X, X, -Y]$	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[11,-22]`
12	$[-X, X, Y]$	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[12,-24]`

Table 21: Wyckoff bond: `24f@12d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[5,-8]`
4	$[Y,\ 0,\ -X]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[6,-7]`
5	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[9,-12]`
6	$[-X,\ Y,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[10,-11]`
7	$[X,\ 0,\ -Y]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[13,-16]`
8	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[14,-15]`
9	$[0,\ Y,\ -X]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[17,-20]`
10	$[0,\ Y,\ X]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[18,-19]`
11	$[Y,\ X,\ 0]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[21,-24]`
12	$[Y,\ -X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[22,-23]`

Table 22: Wyckoff bond: `48g@12d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[17]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[18]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[19]`
20	$[X,\ -Z,\ Y]$	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[20]`
21	$[Z,\ Y,\ -X]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[21]`
22	$[Z,\ -Y,\ X]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[22]`
23	$[-Z,\ Y,\ X]$	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[23]`
24	$[-Z,\ -Y,\ -X]$	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[24]`

* Wyckoff site: `16e`, site symmetry: `.3.`

Table 23: Wyckoff bond: `16a@16e`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$[x,\ x,\ x]$	`[1,5,9]`
2	$[-X,\ -X,\ X]$	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	`[2,7,12]`
3	$[-X,\ X,\ -X]$	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[3,8,10]`
4	$[X,\ -X,\ -X]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	`[4,6,11]`
5	$[X,\ X,\ -X]$	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-x\right]$	`[13,17,21]`
6	$[-X,\ -X,\ -X]$	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	`[14,19,24]`
7	$[X,\ -X,\ X]$	$\left[x+\frac{1}{4},\ \frac{1}{4}-x,\ x+\frac{3}{4}\right]$	`[15,20,22]`
8	$[-X,\ X,\ X]$	$\left[\frac{1}{4}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[16,18,23]`

Table 24: Wyckoff bond: `48b@16e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ x,\ x]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$[x,\ x,\ x]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$[x,\ x,\ x]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-x\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[x+\frac{1}{4},\ \frac{1}{4}-x,\ x+\frac{3}{4}\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-x\right]$	`[17]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{1}{4}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[18]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	`[19]`
20	$[X,\ -Z,\ Y]$	$\left[x+\frac{1}{4},\ \frac{1}{4}-x,\ x+\frac{3}{4}\right]$	`[20]`
21	$[Z,\ Y,\ -X]$	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-x\right]$	`[21]`
22	$[Z,\ -Y,\ X]$	$\left[x+\frac{1}{4},\ \frac{1}{4}-x,\ x+\frac{3}{4}\right]$	`[22]`
23	$[-Z,\ Y,\ X]$	$\left[\frac{1}{4}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[23]`
24	$[-Z,\ -Y,\ -X]$	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	`[24]`

* Wyckoff site: `24f`, site symmetry: `2..`

Table 25: Wyckoff bond: `24a@24f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$\left[\frac{1}{4},\ x,\ 0\right]$	`[5,-8]`

continued ...

Table 25

No.	vector	center	mapping
4	$[Y, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{2} - x, 0\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[0, \frac{1}{4}, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2} - x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, x + \frac{1}{4}, 0\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{3}{4}, \frac{3}{4} - x, \frac{1}{2}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[x + \frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{1}{4} - x, 0, \frac{1}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{4} - x\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, x + \frac{3}{4}\right]$	`[22,-23]`

Table 26: Wyckoff bond: `24b@24f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2,3]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x, 0\right]$	`[5,8]`
4	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{2} - x, 0\right]$	`[6,7]`
5	$[0, 0, X]$	$\left[0, \frac{1}{4}, x\right]$	`[9,12]`
6	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2} - x\right]$	`[10,11]`
7	$[0, X, 0]$	$\left[\frac{3}{4}, x + \frac{1}{4}, 0\right]$	`[13,16]`
8	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4} - x, \frac{1}{2}\right]$	`[14,15]`
9	$[X, 0, 0]$	$\left[x + \frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[17,20]`
10	$[-X, 0, 0]$	$\left[\frac{1}{4} - x, 0, \frac{1}{4}\right]$	`[18,19]`
11	$[0, 0, -X]$	$\left[0, \frac{1}{4}, \frac{1}{4} - x\right]$	`[21,24]`
12	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, x + \frac{3}{4}\right]$	`[22,23]`

Table 27: Wyckoff bond: `48c@24f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, 0, \frac{3}{4}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{4}, x, 0\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{2} - x, 0\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{2} - x, 0\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{4}, x, 0\right]$	`[8]`
9	$[Y, Z, X]$	$\left[0, \frac{1}{4}, x\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2} - x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2} - x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{4}, x\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, x + \frac{1}{4}, 0\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4} - x, \frac{1}{2}\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{3}{4} - x, \frac{1}{2}\right]$	`[15]`

continued ...

Table 27

No.	vector	center	mapping
16	$[-Y,\ X,\ Z]$	$\left[\frac{3}{4},\ x+\frac{1}{4},\ 0\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[x+\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[17]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{1}{4}-x,\ 0,\ \frac{1}{4}\right]$	`[18]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{1}{4}-x,\ 0,\ \frac{1}{4}\right]$	`[19]`
20	$[X,\ -Z,\ Y]$	$\left[x+\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[20]`
21	$[Z,\ Y,\ -X]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-x\right]$	`[21]`
22	$[Z,\ -Y,\ X]$	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{3}{4}\right]$	`[22]`
23	$[-Z,\ Y,\ X]$	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{3}{4}\right]$	`[23]`
24	$[-Z,\ -Y,\ -X]$	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-x\right]$	`[24]`

* Wyckoff site: `24g`, site symmetry: `..2`

Table 28: Wyckoff bond: `24a@24g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y,\ X,\ -X]$	$\left[\frac{1}{8},\ y,\ y+\frac{1}{4}\right]$	`[1,-18]`
2	$[-Y,\ -X,\ -X]$	$\left[\frac{3}{8},\ -y,\ y+\frac{3}{4}\right]$	`[2,-20]`
3	$[-Y,\ X,\ X]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$	`[3,-17]`
4	$[Y,\ -X,\ X]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	`[4,-19]`
5	$[-X,\ Y,\ X]$	$\left[y+\frac{1}{4},\ \frac{1}{8},\ y\right]$	`[5,-15]`
6	$[-X,\ -Y,\ -X]$	$\left[y+\frac{3}{4},\ \frac{3}{8},\ -y\right]$	`[6,-13]`
7	$[X,\ -Y,\ X]$	$\left[\frac{1}{4}-y,\ \frac{7}{8},\ y+\frac{1}{2}\right]$	`[7,-16]`
8	$[X,\ Y,\ -X]$	$\left[\frac{3}{4}-y,\ \frac{5}{8},\ \frac{1}{2}-y\right]$	`[8,-14]`
9	$[X,\ -X,\ Y]$	$\left[y,\ y+\frac{1}{4},\ \frac{1}{8}\right]$	`[9,-21]`
10	$[-X,\ -X,\ -Y]$	$\left[-y,\ y+\frac{3}{4},\ \frac{3}{8}\right]$	`[10,-23]`
11	$[X,\ X,\ -Y]$	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{7}{8}\right]$	`[11,-22]`
12	$[-X,\ X,\ Y]$	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{5}{8}\right]$	`[12,-24]`

Table 29: Wyckoff bond: `24b@24g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ X,\ X]$	$\left[\frac{1}{8},\ y,\ y+\frac{1}{4}\right]$	`[1,18]`
2	$[0,\ -X,\ X]$	$\left[\frac{3}{8},\ -y,\ y+\frac{3}{4}\right]$	`[2,20]`
3	$[0,\ X,\ -X]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$	`[3,17]`
4	$[0,\ -X,\ -X]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	`[4,19]`
5	$[X,\ 0,\ X]$	$\left[y+\frac{1}{4},\ \frac{1}{8},\ y\right]$	`[5,15]`
6	$[X,\ 0,\ -X]$	$\left[y+\frac{3}{4},\ \frac{3}{8},\ -y\right]$	`[6,13]`
7	$[-X,\ 0,\ X]$	$\left[\frac{1}{4}-y,\ \frac{7}{8},\ y+\frac{1}{2}\right]$	`[7,16]`
8	$[-X,\ 0,\ -X]$	$\left[\frac{3}{4}-y,\ \frac{5}{8},\ \frac{1}{2}-y\right]$	`[8,14]`
9	$[X,\ X,\ 0]$	$\left[y,\ y+\frac{1}{4},\ \frac{1}{8}\right]$	`[9,21]`
10	$[-X,\ X,\ 0]$	$\left[-y,\ y+\frac{3}{4},\ \frac{3}{8}\right]$	`[10,23]`
11	$[X,\ -X,\ 0]$	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{7}{8}\right]$	`[11,22]`
12	$[-X,\ -X,\ 0]$	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{5}{8}\right]$	`[12,24]`

Table 30: Wyckoff bond: `48c@24g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{8}, y, y+\frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{8}, -y, y+\frac{3}{4}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{7}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{5}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	`[4]`
5	$[Z, X, Y]$	$\left[y+\frac{1}{4}, \frac{1}{8}, y\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[y+\frac{3}{4}, \frac{3}{8}, -y\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{4}-y, \frac{7}{8}, y+\frac{1}{2}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}-y, \frac{5}{8}, \frac{1}{2}-y\right]$	`[8]`
9	$[Y, Z, X]$	$\left[y, y+\frac{1}{4}, \frac{1}{8}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[-y, y+\frac{3}{4}, \frac{3}{8}\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{7}{8}\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{5}{8}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[y+\frac{3}{4}, \frac{3}{8}, -y\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4}-y, \frac{5}{8}, \frac{1}{2}-y\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{8}, y\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{1}{4}-y, \frac{7}{8}, y+\frac{1}{2}\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[\frac{7}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{8}, y, y+\frac{1}{4}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{5}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[\frac{3}{8}, -y, y+\frac{3}{4}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[y, y+\frac{1}{4}, \frac{1}{8}\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{7}{8}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[-y, y+\frac{3}{4}, \frac{3}{8}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{5}{8}\right]$	`[24]`

* Wyckoff site: `24h`, site symmetry: `..2`

Table 31: Wyckoff bond: `24a@24h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	`[1,-19]`
2	$[-Y, -X, X]$	$\left[\frac{3}{8}, -y, \frac{3}{4}-y\right]$	`[2,-17]`
3	$[-Y, X, -X]$	$\left[\frac{7}{8}, y+\frac{1}{2}, y+\frac{1}{4}\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[\frac{5}{8}, \frac{1}{2}-y, y+\frac{3}{4}\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[\frac{3}{4}-y, \frac{3}{8}, -y\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[y+\frac{1}{4}, \frac{7}{8}, y+\frac{1}{2}\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[y+\frac{3}{4}, \frac{5}{8}, \frac{1}{2}-y\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[-y, \frac{3}{4}-y, \frac{3}{8}\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[y+\frac{1}{2}, y+\frac{1}{4}, \frac{7}{8}\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[\frac{1}{2}-y, y+\frac{3}{4}, \frac{5}{8}\right]$	`[12,-21]`

Table 32: Wyckoff bond: `24b@24h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ X,\ -X]$	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	`[1,19]`
2	$[0,\ -X,\ -X]$	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	`[2,17]`
3	$[0,\ X,\ X]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	`[3,20]`
4	$[0,\ -X,\ X]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	`[4,18]`
5	$[-X,\ 0,\ X]$	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	`[5,14]`
6	$[-X,\ 0,\ -X]$	$\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$	`[6,16]`
7	$[X,\ 0,\ X]$	$\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$	`[7,13]`
8	$[X,\ 0,\ -X]$	$\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$	`[8,15]`
9	$[X,\ -X,\ 0]$	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	`[9,24]`
10	$[-X,\ -X,\ 0]$	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$	`[10,22]`
11	$[X,\ X,\ 0]$	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$	`[11,23]`
12	$[-X,\ X,\ 0]$	$\left[\frac{1}{2}-y,\ y+\frac{3}{4},\ \frac{5}{8}\right]$	`[12,21]`

Table 33: Wyckoff bond: `48c@24h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2}-y,\ y+\frac{3}{4},\ \frac{5}{8}\right]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[y+\frac{1}{4},\ \frac{7}{8},\ y+\frac{1}{2}\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4}-y,\ \frac{1}{8},\ y\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[y+\frac{3}{4},\ \frac{5}{8},\ \frac{1}{2}-y\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[\frac{3}{4}-y,\ \frac{3}{8},\ -y\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	`[17]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ y+\frac{3}{4}\right]$	`[18]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	`[19]`
20	$[X,\ -Z,\ Y]$	$\left[\frac{7}{8},\ y+\frac{1}{2},\ y+\frac{1}{4}\right]$	`[20]`
21	$[Z,\ Y,\ -X]$	$\left[\frac{1}{2}-y,\ y+\frac{3}{4},\ \frac{5}{8}\right]$	`[21]`
22	$[Z,\ -Y,\ X]$	$\left[-y,\ \frac{3}{4}-y,\ \frac{3}{8}\right]$	`[22]`
23	$[-Z,\ Y,\ X]$	$\left[y+\frac{1}{2},\ y+\frac{1}{4},\ \frac{7}{8}\right]$	`[23]`
24	$[-Z,\ -Y,\ -X]$	$\left[y,\ \frac{1}{4}-y,\ \frac{1}{8}\right]$	`[24]`

* Wyckoff site: `48i`, site symmetry: `1`

Table 34: Wyckoff bond: `48a@48i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, -y, z + \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, y + \frac{1}{2}, \frac{1}{2} - z\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - y, -z\right]$	`[4]`
5	$[Z, X, Y]$	$[z, x, y]$	`[5]`
6	$[Z, -X, -Y]$	$\left[z + \frac{1}{2}, \frac{1}{2} - x, -y\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2} - z, -x, y + \frac{1}{2}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[-z, x + \frac{1}{2}, \frac{1}{2} - y\right]$	`[8]`
9	$[Y, Z, X]$	$[y, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$\left[-y, z + \frac{1}{2}, \frac{1}{2} - x\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[y + \frac{1}{2}, \frac{1}{2} - z, -x\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2} - y, -z, x + \frac{1}{2}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[y + \frac{3}{4}, x + \frac{1}{4}, \frac{1}{4} - z\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{3}{4} - y, \frac{3}{4} - x, \frac{3}{4} - z\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[y + \frac{1}{4}, \frac{1}{4} - x, z + \frac{3}{4}\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{1}{4} - y, x + \frac{3}{4}, z + \frac{1}{4}\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[x + \frac{3}{4}, z + \frac{1}{4}, \frac{1}{4} - y\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{4} - x, z + \frac{3}{4}, y + \frac{1}{4}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{3}{4} - x, \frac{3}{4} - z, \frac{3}{4} - y\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[x + \frac{1}{4}, \frac{1}{4} - z, y + \frac{3}{4}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[z + \frac{3}{4}, y + \frac{1}{4}, \frac{1}{4} - x\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[z + \frac{1}{4}, \frac{1}{4} - y, x + \frac{3}{4}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{4} - z, y + \frac{3}{4}, x + \frac{1}{4}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{3}{4} - z, \frac{3}{4} - y, \frac{3}{4} - x\right]$	`[24]`